

RUDRAKSH MALI

Loughborough, UK · +44 7554469356 · rudrakshmali1@gmail.com · <https://rudramali.github.io/rudrakshmali-portfolio/>

PROFILE

Mechanical engineer with hands-on experience across structural design, FEA analysis, CAD modelling, DFM, and R&D product development. Currently completing an MSc in Engineering Management at Loughborough University and gaining commercial NPD experience as a Product Design Engineer intern at a Loughborough University startup. Proven track record of leading multidisciplinary teams, delivering engineering programmes under real constraints, and producing solutions that hold up in physical reality. Seeking graduate or junior engineering roles in design, R&D, manufacturing, or technical programme management.

EDUCATION

MSc – Engineering Management | Loughborough University Oct 2025 – Present · UK

- Relevant modules: Systems Engineering, Engineering Project Management, Operations Management, Sustainable Engineering, Manufacturing Process & Automation, Product Development, Corporate Finance, Business & Entrepreneurship, People Organisation & Leadership.
- Dissertation: Designing a cloud-based additive manufacturing platform ecosystem — stakeholder architecture, AMaaS service modelling, and distributed manufacturing resource optimisation across complex multi-party networks.

BE – Mechanical Engineering | Rajiv Gandhi Institute of Technology, Mumbai University Aug 2021 – May 2025 · India

- Relevant modules: Additive Manufacturing, Smart Materials, Energy Harvesting, Manufacturing Technology, Structural Analysis.

EXPERIENCE

Product Design Engineer (Intern) | Amora Collective – Loughborough University Startup Jun 2026 – Present · Loughborough, UK

- Supporting DFM and design development of a consumer health product being taken from prototype to commercial launch — responsibilities include CAD design, material review, supplier discussions, and design changes for manufacturability.
- Working within a fast-moving startup environment backed by Cambridge Consultants, PriestmanGoode, and P&G, gaining hands-on experience of commercial NPD decision-making from design through to production readiness.

Team Captain & Lead Design Engineer | Team RGIT Racing – Formula Student Aug 2021 – Feb 2024 · Mumbai, India

- Led a 25-member multidisciplinary engineering team through full design, build, and validation of a Formula Student race car — coordinating mechanical, structural, and manufacturing workstreams across a multi-year programme.
- Redesigned the chassis from 40 kg to 31 kg (22.5% weight reduction) through systematic FEA-based structural optimisation in SolidWorks and ANSYS, balancing torsional rigidity, manufacturability, and cost targets.
- Made a strategic decision to reuse the existing chassis architecture to preserve team budget (Rs 13 lakhs, ~£12,000), then redesigned every surrounding system to hit performance targets — delivering impact under real financial constraints.
- Collaborated with electronics team on safety-critical vehicle systems integration including APS, brake/crush sensors, and shutdown circuits — developing practical understanding of mechanical-electronic system interactions.
- Secured All India Rank 9 at Formula Bharat 2023 — validating engineering decisions made under significant time, budget, and resource constraints.

R&D Design Intern | Go Float Tech Jan 2024 – Jul 2024 · India

- Owned the full CAD design of an underwater drone prototype; the majority of designed components were adopted into the final production unit.
- Proposed and implemented design modifications beyond the original scope that were incorporated into the official product, demonstrating technical credibility and initiative in a professional R&D environment.

- Gained end-to-end exposure to a product development lifecycle: concept, iterative design, materials selection, testing, and assembly validation.

KEY PROJECTS

Cloud-Based Additive Manufacturing Platform · MSc Dissertation (Ongoing)

- Designing a digital service ecosystem connecting customers, AM service providers, material suppliers, and logistics partners — applying systems architecture and stakeholder modelling to a real gap in how manufacturing capacity is accessed and shared globally.

Multi-Directional Rotating Loading Bay · Bachelor's Final Year Project · 2024

- Designed and fabricated a novel multi-directional rotation mechanism for logistics material handling — full 3D CAD modelling, FEA structural validation, and physical prototype build.

Formula Student Chassis Structural Optimisation · 2022–2023

- FEA-driven redesign targeting stiffness-to-weight improvement; implemented torsional rigidity enhancements through iterative CAD modelling, simulation, and design review cycles.

TECHNICAL SKILLS

Design & Simulation: SolidWorks (advanced), ANSYS (FEA & structural), AutoCAD, MATLAB

Manufacturing: Additive Manufacturing, CNC Machining, Prototyping, Smart Materials, DFM, DFA

Engineering Methods: Structural optimisation, FEA validation, Systems engineering, Design for manufacture

Project & Leadership: Programme delivery, Budget management, Team leadership (25 members), Design reviews

Digital & Data: Python (basic), MS Excel, Data analysis, Technical documentation

ACHIEVEMENTS & INTERESTS

- All India Rank 9 — Formula Bharat 2023 (national Formula Student competition).
- Team Captain for 1 year — led RGIT Racing through design, build, and national competition at Formula Bharat 2023.
- Strong interest in advanced manufacturing, motorsport engineering, Industry 4.0, and the future of distributed production systems.
- Right to work in the UK — valid visa, eligible from summer 2026.